

HYDROGENATED NBR SYNTHESIS AND LOW-MOLECULAR-WEIGHT HNBR PROCESS LANDSCAPE

1. ABSTRACT

This report synthesizes eight patents focused on hydrogenated nitrile rubber (HNBR) synthesis, precursor preparation, and post-hydrogenation purification. The target material is hydrogenated or partially hydrogenated nitrile rubber derived from NBR, typically an acrylonitrile/butadiene copolymer optionally containing additional comonomers. Across the patent set, the central technical objective is to reduce residual unsaturation while controlling molecular weight, Mooney viscosity, polydispersity, and catalyst residues so that HNBR retains heat, ozone, oil, and aging resistance with improved processability. The landscape divides into three major themes: direct hydrogenation of NBR in specialized reactors, one-step or simultaneous metathesis/hydrogenation routes to low-molecular-weight HNBR, and downstream catalyst-removal methods for improving aging performance and color.

US7666950B2 discloses hydrogenation of NBR in a spinning disk reactor, emphasizing lower hydrogen pressure than conventional batch systems and use of homogeneous rhodium/ruthenium catalysts. US20210340285A1 focuses on solution hydrogenation of partially unsaturated NBR using Ru, Pd, or Rh compounds at 60-200°C and 0.7-15 MPa, with explicit attention to minimizing residual catalyst and improving aging. EP2473281B1 and US9150669B2 are the most process-intensive low-MW HNBR disclosures: both describe simultaneous metathesis and hydrogenation in the presence of hydrogen and ruthenium/osmium catalysts, enabling single-step reduction of molecular weight and residual double bonds. US8362154B2 and US7951875B2 are precursor-degradation patents that reduce NBR molecular weight by metathesis before or during hydrogenation, using Grubbs-type catalysts, co-olefins such as 1-hexene or ethylene, and controlled catalyst loadings. US9023914B2 provides a complementary route by preparing NBR in organic solvent with modifiers, enabling tailored precursor architecture before optional hydrogenation. US7846995B2 addresses the purification problem by removing Ru residues from optionally hydrogenated NBR using functionalized ion-exchange resins.

Overall, the evidence supports solution-phase hydrogenation or combined metathesis/hydrogenation as the dominant routes, with catalyst choice, solvent, and post-treatment being key levers for achieving low-RDB, low-MW HNBR suitable for demanding elastomer applications.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7666950B2
- Patent Title: Process for preparing hydrogenated nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparing hydrogenated nitrile rubber by subjecting NBR to hydrogenation in a spinning disk reactor. The process is presented as less demanding in hydrogen pressure than conventional batch hydrogenation and is intended to reduce equipment and manufacturing cost. - Hydrogen pressure: 3.5 bar to 100 bar — Claim 1 - Hydrogen pressure: 65 to 90 bar — Background discussion of batch hydrogenation - Mooney viscosity of commercially available HNBR: 55 to 120 ML 1+4 @100°C — Background discussion - Residual double bond content: up to 18% — Background discussion - Mooney viscosity of NBR: 30 to 90 ML 1+4 @100°C — Detailed description - Mooney viscosity of NBR: 30 to 70 ML 1+4 @100°C — Detailed description - Mooney viscosity of NBR: 30 to 50 ML 1+4 @100°C — Detailed description - Molecular weight Mw of NBR: 200,000 to 500,000 — Detailed description - Molecular weight Mw of NBR: 200,000 to 400,000 — Detailed description - Molecular weight Mn of NBR: 80,000 to 150,000 — Detailed description - Polydispersity index PDI of NBR: 2.0 to 6.0 — Detailed description - Polydispersity index PDI of NBR: 2.0 to 4.0 — Detailed description - NBR composition: 40 to 90 wt.% conjugated diene — Detailed description - NBR composition: 60 to 85 wt.% conjugated diene — Detailed description - NBR composition: 10 to 60 wt.% α,β -unsaturated nitrile — Detailed description - NBR composition: 15 to 40 wt.% α,β -unsaturated nitrile — Detailed description - Additional monomers: 0 to 40 wt.% — Detailed description - Additional monomers: 0.1 to 40 wt.% — Detailed description - Additional monomers: 1 to 30 wt.% — Detailed description

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is directly relevant because it targets the core HNBR hydrogenation step from NBR and provides a reactor-based alternative to conventional batch processing. It is especially important for process intensification and pressure reduction in HNBR manufacture.

Patent 2

- Patent Number: US8362154B2
- Patent Title: Metathesis of nitrile rubbers in the presence of transition metal complex ...

- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses molecular-weight degradation of NBR by metathesis using ruthenium/osmium catalysts, with the degraded rubber then serving as a lower-MW precursor for HNBR production. The examples compare Grubbs III and Grubbs II catalysts and show that the newer catalyst gives faster degradation and narrower molecular-weight distributions. - NBR amount: 40 g — Example 1 - Grubbs III catalyst amount: 80 mg — Example 1 - Grubbs III catalyst loading: 0.02 phr — Example 1 - Ruthenium content: 23 ppm — Example 1 - 1-Hexene amount: 0.8 g — Example 1 - 1-Hexene loading: 2.0 phr — Example 1 - Temperature: 23 °C — Example 1 - Reaction time: 0, 15, 60 min — Example 1 - Mw: 240 kg/mol at 0 min — Example 1 - Mw: 150 kg/mol at 15 min — Example 1 - Mw: 144 kg/mol at 60 min — Example 1 - Mn: 100 kg/mol at 0 min — Example 1 - Mn: 71 kg/mol at 15 min — Example 1 - Mn: 72 kg/mol at 60 min — Example 1 - PDI: 2.4 at 0 min — Example 1 - PDI: 2.1 at 15 min — Example 1 - PDI: 2.0 at 60 min — Example 1 - NBR amount: 40 g — Example 2 - Grubbs III catalyst amount: 200 mg — Example 2 - Grubbs III catalyst loading: 0.05 phr — Example 2 - Ruthenium content: 57 ppm — Example 2 - 1-Hexene amount: 0.8 g — Example 2 - 1-Hexene loading: 2.0 phr — Example 2 - Temperature: 23 °C — Example 2 - Reaction time: 0, 15, 90 min — Example 2 - Mw: 240 kg/mol at 0 min — Example 2 - Mw: 39 kg/mol at 15 min — Example 2 - Mw: 36 kg/mol at 90 min — Example 2 - Mn: 100 kg/mol at 0 min — Example 2 - Mn: 21 kg/mol at 15 min — Example 2 - Mn: 20 kg/mol at 90 min — Example 2 - PDI: 2.4 at 0 min — Example 2 - PDI: 1.8 at 15 min — Example 2 - PDI: 1.8 at 90 min — Example 2

Relevant Experimental Evidence - Example 1: Grubbs III catalyst at 23 ppm Ru with 2.0 phr 1-hexene reduced Mw from 240 to 144 kg/mol and gave gel-free degraded NBR. - Example 1: Grubbs II catalyst under the same Ru and co-olefin loading was less active and gave higher Mw and broader PDI. - Example 2: Increasing Ru loading to 57 ppm with Grubbs III catalyst drove Mw down to 36 kg/mol and PDI to 1.8. - Example 2: Grubbs II catalyst at 57 ppm Ru was less effective than Grubbs III at identical conditions. - Example 2: Both examples demonstrate that catalyst structure strongly controls metathesis rate and final precursor molecular weight.

Technical Relevance This patent is relevant because it provides a route to low-molecular-weight NBR precursors that can be hydrogenated to low-MW HNBR. It establishes catalyst and co-olefin conditions that directly influence the final HNBR molecular-weight profile.

Patent 3

- Patent Number: US7951875B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparing low molecular weight hydrogenated nitrile rubber by first dissolving NBR in monochlorobenzene and then carrying out metathesis/degradation in the presence of a ruthenium catalyst and co-olefin, followed by hydrogenation. The examples show that catalyst loading, co-olefin choice, and reaction time strongly affect Mw, Mn, PDI, and Mooney viscosity. - Rubber amount: 200 g — Example 1 - MCB amount: 1133 g — Example 1 - Solid content: 15 wt.-% — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Rubber amount: 200 g — Example 2 - MCB amount: 1133 g — Example 2 - Solid content: 15 wt.-% — Example 2 - Ethylene pressure: 6.9×10^5 Pa — Example 2 - Rubber amount: 75 g — Example 3 - MCB amount: 1175 g — Example 3 - Solid content: 6 wt.-% — Example 3 - Ethylene pressure: 6.9×10^5 Pa — Example 3 - Rubber amount: 75 g — Example 4 - MCB amount: 1175 g — Example 4 - Solid content: 6 wt.-% — Example 4 - 1-Hexene amount: 150 g — Example 4 - Temperature: 80 °C — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Co-olefin concentration: 500 psi — Table 1 - Co-olefin amount: 150 g — Table 1 - Mw of starting polymer: 296 kg/mol — Table 2 - Mn of starting polymer: 85 kg/mol — Table 2 - PDI of starting polymer: 3.50 — Table 2 - Mooney viscosity of starting polymer: 73 — Table 2 - Mn: 27 kg/mol — Table 2, typical product - Mw: 54 kg/mol — Table 2, typical product - PDI: 2.0 — Table 2, typical product - Mw: 189 kg/mol — Experiment 1 - Mn: 73 kg/mol — Experiment 1 - PDI: 2.59 — Experiment 1 - Mooney viscosity: 43 — Experiment 1 - Mw: 136 kg/mol — Experiment 2 - Mn: 60 kg/mol — Experiment 2 - PDI: 2.27 — Experiment 2 - Mooney viscosity: 28 — Experiment 2 - Mw: 59 kg/mol — Experiment 3 - Mn: 31 kg/mol — Experiment 3 - PDI: 1.90 — Experiment 3 - Mooney viscosity: 3 — Experiment 3 - Mw: 111 kg/mol — Experiment 4 - Mn: 55 kg/mol — Experiment 4 - PDI: 2.02 — Experiment 4 - Mooney viscosity: 31 — Experiment 4

Relevant Experimental Evidence - Example 1: Ethylene-assisted metathesis at 80 °C and 500 psi co-olefin gave a lower-MW HNBR product. - Example 2: Higher catalyst loading produced further molecular-weight reduction and lower Mooney viscosity. - Example 3: Lower solids content and higher catalyst loading yielded the lowest Mw and PDI in the series. - Example 4: 1-Hexene could replace ethylene as co-olefin and still produce low-MW HNBR. - Example 4: The data show that metathesis conditions can be tuned to obtain HNBR with Mooney viscosity as low as 3.

Technical Relevance This patent is highly relevant because it directly discloses a practical route to low-MW HNBR through metathesis followed by hydrogenation. It provides concrete process windows for solvent, co-olefin, catalyst loading, and temperature that govern final HNBR properties.

Patent 4

- Patent Number: US20210340285A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses hydrogenating at least partially unsaturated NBR in solution using ruthenium, palladium, or rhodium compounds, with the stated goal of producing HNBR having good aging properties and low residual catalyst. The disclosure emphasizes catalyst recovery/removal and the adverse effects of noble-metal residues on color, gel content, and aging. - Ruthenium amount: 10 ppm to 200 ppm — Claim 1 - Ruthenium amount: 10 ppm to 150 ppm — Claim 1 - Ruthenium amount: 10 ppm to 120 ppm — Claim 1 - Ruthenium amount: 10 ppm to 79 ppm — Claim 1 - Ruthenium amount: 42 ppm to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm — Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200 °C — Claim 1 - Hydrogenation pressure: 700000 to 15000000 Pa — Claim 1 - Hydrogenation time: 1 to 24 h — Claim 1 - HNBR composition: 40 to 90 wt.% conjugated diene — Claim 1 - HNBR composition: 10 to 60 wt.% α,β -unsaturated nitrile — Claim 1 - Rhodium content in cited recovery example: 58 to 60 ppm — Background discussion - Removal rate: 99% — Background discussion of tin-containing complexing reagent method

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is relevant because it addresses the direct hydrogenation step for HNBR and the critical issue of catalyst residue control. Its disclosed catalyst loadings and reaction windows are useful for balancing hydrogenation efficiency with product aging performance.

Patent 5

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-step process in which NBR is simultaneously subjected to metathesis and hydrogenation in the presence of hydrogen and a ruthenium- or osmium-based catalyst. The process is designed to produce low-molecular-weight HNBR with reduced residual double bonds, lower Mooney viscosity, and narrower molecular-weight distribution than conventional two-step routes. - Molecular weight Mw of HNBR: 20,000 to 250,000 — Description - Mooney viscosity: 1 to 50 ML 1+4 @100°C — Description - Polydispersity index: less than 3.0 — Description - NBR Mooney viscosity: 24 to 70 ML 1+4 at 100°C — Description - NBR Mooney viscosity: 28 to 40 ML 1+4 at 100°C — Description - NBR molecular weight Mw: 200,000 to 500,000 — Description - NBR molecular weight Mw: 200,000 to 400,000 — Description - NBR composition: 40 to 90 wt.% conjugated diene — Description - NBR composition: 60 to 85 wt.% conjugated diene — Description - NBR composition: 10 to 60 wt.% α,β -unsaturated nitrile — Description - NBR composition: 15 to 40 wt.% α,β -unsaturated nitrile — Description - Additional monomers: 0 to 40 wt.% — Description - Additional monomers: 0.1 to 40 wt.% — Description - Additional monomers: 1 to 30 wt.% — Description

Relevant Experimental Evidence - Example 1: Hoveyda catalyst at 0.041 phr and 65.6 ppm metal gave faster hydrogenation than Wilkinson's catalyst. - Example 1: HNBR 1 reached RDB 0.4% after 240 min and had Mw 113,371 g/mol and Mooney 33.3. - Example 1: HNBR 2 with Arlt catalyst showed intermediate hydrogenation performance and Mw 133,758 g/mol. - Example 1: Wilkinson's catalyst with triphenylphosphine co-catalyst gave the highest final Mw and Mooney viscosity among the three systems. - Example 1: The data demonstrate that catalyst identity strongly affects hydrogenation rate and final HNBR molecular weight.

Technical Relevance This patent is one of the most directly relevant disclosures because it combines metathesis and hydrogenation to make low-MW HNBR in a single step. It provides a clear process target for reducing both residual unsaturation and viscosity while controlling molecular-weight distribution.

Patent 6

- Patent Number: US9023914B2
- Patent Title: Nitrile rubbers and production thereof in organic solvents
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses free-radical polymerization of NBR in organic solvent using specific modifiers, with optional subsequent hydrogenation to HNBR. The process is aimed at producing nitrile rubbers with tailored molecular weights, polydispersities, and structural elements from the modifier in the main chain or chain

ends. - Acrylonitrile: 37 phm — Example 1 / Table 1 - 1,3-Butadiene: 63 phm — Example 1 / Table 1 - Dimethylacetamide: 51 phm — Example 1 / Table 1 - Vazo® 88: 0.310 phm — Example 1 / Table 1 - DoPAT: 0.222 phm — Example 1 / Table 1 - Polymerization temperature: 100 °C — Example 1 / Table 1 - Polymerization time: 9 h — Example 1 / Table 1 - Final conversion: 50% — Example 1 / Table 1 - Hydroxyethyl acrylate: 20 phm — Example 6 / Table 1 - Acrylonitrile: 30 phm — Example 6 / Table 1 - 1,3-Butadiene: 50 phm — Example 6 / Table 1 - Polymerization time: 6 h — Example 6 / Table 1 - Final conversion: 39% — Example 6 / Table 1 - DoPAT-NBR#7: 50 phm — Example 8 / Table 1 - Styrene: 50 phm — Example 8 / Table 1 - Polymerization temperature: 100 °C — Example 8 / Table 1 - Polymerization time: 13 h — Example 8 / Table 1 - Final conversion: 9.2% — Example 8 / Table 1 - NBR #7 Mn: 12,000 g/mol — Example 8 - DoPAT: 2.22 phm — Example 7 / Table 1 - Polymerization time: 9 h — Example 7 / Table 1 - Final conversion: 48% — Example 7 / Table 1 - MCB: 56 phm — Example 3 / Table 1 - MCB: 56 phm — Example 4 / Table 1 - DMAc: 202 phm — Example 5 / Table 1 - Vazo® 88: 0.23 phm — Example 5 / Table 1 - Polymerization time: 5 h — Example 5 / Table 1 - Final conversion: 28% — Example 5 / Table 1

Relevant Experimental Evidence - Example 1: Organic-solvent NBR synthesis at 100 °C with acrylonitrile/butadiene feed and modifier/initiator system reached 50% conversion. - Example 6: Hydroxyethyl acrylate monomer could be incorporated into NBR in organic solvent. - Example 7: Increasing modifier loading produced a deliberately low-molecular-mass NBR precursor. - Example 8: The low-Mn NBR precursor from Example 7 was used as a RAFT macromer for block-styrene extension. - Example 1-8: The examples show that solvent and modifier choice can tune precursor architecture before optional hydrogenation.

Technical Relevance This patent is relevant as a precursor-manufacturing disclosure that enables controlled NBR feedstocks for later hydrogenation to HNBR. It is especially useful where low molecular weight or functionalized HNBR is desired.

Patent 7

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of NBR in the presence of hydrogen and a ruthenium/osmium catalyst, producing low-molecular-weight HNBR in a single step. The examples compare Hoveyda, Arlt, and Wilkinson-type systems and show that catalyst choice controls hydrogenation rate, residual double bond content, and final molecular weight. - Molecular weight Mw of HNBR: 20,000

to 250,000 — Description - Mooney viscosity: 1 to 50 ML 1+4 at 100°C — Description - Polydispersity index: less than 3.0 — Description - NBR Mooney viscosity: 24 to 70 ML 1+4 at 100°C — Description - NBR Mooney viscosity: 28 to 40 ML 1+4 at 100°C — Description - NBR molecular weight Mw: 200,000 to 500,000 — Description - NBR molecular weight Mw: 200,000 to 400,000 — Description - NBR composition: 40 to 90 wt.% conjugated diene — Description - NBR composition: 60 to 85 wt.% conjugated diene — Description - NBR composition: 10 to 60 wt.% α,β -unsaturated nitrile — Description - NBR composition: 15 to 40 wt.% α,β -unsaturated nitrile — Description - Acrylonitrile: 37 phm — Example 1 / Table 1 - 1,3-Butadiene: 63 phm — Example 1 / Table 1 - Dimethylacetamide: 51 phm — Example 1 / Table 1 - Vazo® 88: 0.310 phm — Example 1 / Table 1 - DoPAT: 0.222 phm — Example 1 / Table 1 - Polymerization temperature: 100 °C — Example 1 / Table 1 - Polymerization time: 9 h — Example 1 / Table 1 - Final conversion: 50% — Example 1 / Table 1 - Hydrogenation catalyst loading: 0.041 phr — Example 1 / Table 1 - Hydrogenation catalyst loading: 0.045 phr — Example 1 / Table 1 - Hydrogenation catalyst loading: 0.060 phr — Example 1 / Table 1 - Metal content: 65.6 ppm — Example 1 / Table 1 - Co-catalyst amount: 1.0 phr triphenylphosphine — Example 1 / Table 1 - Reaction time: 0, 60, 120, 180, 240 min — Example 1 / Table 2 - RDB: 100% at 0 min — Example 1 / Table 2 - RDB: 2.5%, 0.6%, 0.5%, 0.4% for HNBR 1 at 60, 120, 180, 240 min — Example 1 / Table 2 - RDB: 7.0%, 4.5%, 3.9%, 3.5% for HNBR 2 at 60, 120, 180, 240 min — Example 1 / Table 2 - RDB: 15.4%, 3.1%, 1.3%, 0.7% for HNBR 3 at 60, 120, 180, 240 min — Example 1 / Table 2 - Mw: 113371 g/mol — Example 1 / Table 3 - Mn: 48352 g/mol — Example 1 / Table 3 - PDI: 2.34 — Example 1 / Table 3 - Mooney viscosity: 33.3 — Example 1 / Table 3 - Mw: 133758 g/mol — Example 1 / Table 3 - Mn: 54595 g/mol — Example 1 / Table 3 - PDI: 2.45 — Example 1 / Table 3 - Mooney viscosity: 41.3 — Example 1 / Table 3 - Mw: 195304 g/mol — Example 1 / Table 3 - Mn: 64130 g/mol — Example 1 / Table 3 - PDI: 3.05 — Example 1 / Table 3 - Mooney viscosity: 69.3 — Example 1 / Table 3

Relevant Experimental Evidence - Example 1: Hoveyda catalyst at 0.041 phr and 65.6 ppm metal gave the fastest hydrogenation profile. - Example 1: HNBR 1 reached 0.4% RDB after 240 min and had the lowest Mooney viscosity. - Example 1: Arlt catalyst gave intermediate hydrogenation performance and intermediate molecular weight. - Example 1: Wilkinson's catalyst with triphenylphosphine co-catalyst gave the slowest hydrogenation and highest final Mooney viscosity. - Example 1: The data confirm that catalyst system selection governs both hydrogenation efficiency and final HNBR rheology.

Technical Relevance This patent is highly relevant because it directly claims a one-step route to low-MW HNBR from NBR under hydrogen. It provides a practical catalyst comparison showing how to achieve low residual unsaturation and lower viscosity in the final product.

Patent 8

- Patent Number: US7846995B2
- Patent Title: ... -containing catalyst residues from optionally hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US

- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses removing ruthenium-containing catalyst residues from optionally hydrogenated nitrile rubber solutions using functionalized ion-exchange resins. The process is intended to improve aging properties, color, and product quality by lowering residual Ru after hydrogenation or metathesis-related processing. - Ruthenium in feed solution: 5 to 1000 ppm — Claim 2 - Ruthenium in feed solution: 5 to 500 ppm — Claim 3 - HNBR concentration in solution: 0.5 to 30 wt.% — Claim 4 - Resin particle diameter: 0.2 to 2.5 mm dry basis — Claim 1 - Treatment temperature: ca. 100 °C — Example 2A - Treatment time: 66 h — Example 2A - Rubber solution amount: 180 g — Example 2A - Ion exchange resin amount: 0.5 g — Example 2A - Initial Ru content: 15 ppm — Table 2 / Example 2A and control - Ru content after treatment: 10 ppm — Example 2A - Ru removal: 33% — Example 2A

Relevant Experimental Evidence - Example 2A: Treating a standard nitrile rubber solution with 0.5 g Lewatit® MonoPlus MP 500 at about 100 °C for 66 h reduced Ru to 10 ppm. - Example 2A: The ion-exchange treatment removed 33% of the Ru from the rubber solution. - Example 2B: The untreated control sample established the initial Ru content for comparison. - Example 1: The example block indicates a batchwise purification context, though detailed conditions are not fully captured in the extraction. - Example 2A/2B: The data demonstrate that functionalized resin treatment can lower catalyst residue in rubber solutions.

Technical Relevance This patent is relevant because catalyst residue removal is a critical finishing step for HNBR, especially for aging-sensitive applications. It provides a practical purification route to lower Ru content after hydrogenation or metathesis-based synthesis.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7666950B2 emphasizes reactor intensification via spinning disk hydrogenation, whereas US20210340285A1 and US9150669B2 focus on solution hydrogenation with noble-metal catalysts and explicit residue control.
- US9150669B2 and EP2473281B1 both target low-MW HNBR by simultaneous metathesis/hydrogenation; both use ruthenium/osmium catalyst families, but EP2473281B1 reports Hoveyda/Arlt/Wilkinson comparisons while US9150669B2 frames the process more broadly with formula (I) catalysts.
- US8362154B2 and US7951875B2 are precursor-degradation routes: both use metathesis with co-olefins to lower NBR molecular weight before or during hydrogenation, but US7951875B2 provides more explicit solvent/co-olefin/process data and final HNBR property tables.

- US8362154B2 shows Grubbs III outperforming Grubbs II at 23 and 57 ppm Ru, while US7951875B2 demonstrates that ethylene or 1-hexene and catalyst loading from 0.05 to 0.25 phr can drive Mw down to 54 kg/mol and Mooney viscosity to 3.
- US9023914B2 is upstream of the HNBR step: it prepares tailored NBR in organic solvent with modifiers, enabling controlled precursor architecture before hydrogenation, unlike the other patents that focus on hydrogenation or purification.
- US7846995B2 is downstream of synthesis and addresses Ru removal rather than polymer formation; it complements the hydrogenation patents by improving aging and color through ion-exchange purification.
- Across the hydrogenation patents, the most common catalyst families are rhodium, ruthenium, and palladium compounds, with hydrogenation temperatures spanning 60 to 200 °C and pressures from 3.5 bar up to 15 MPa depending on reactor and catalyst system.
- Residual unsaturation control is strongest in the simultaneous metathesis/hydrogenation patents, where RDB values are driven below 1% in the best examples, while direct hydrogenation patents emphasize catalyst recovery and aging stability.

4. CONCLUSION

The evidence shows that HNBR can be produced by three main routes: direct hydrogenation of NBR, simultaneous metathesis/hydrogenation to lower molecular weight during saturation, and precursor tailoring followed by hydrogenation. For the most process-efficient low-MW HNBR, the strongest evidence supports one-step metathesis/hydrogenation using ruthenium/osmium catalysts in solution, with catalyst loadings in the low-ppm to low-phr range and temperatures around 23 to 80 °C in the examples. For conventional hydrogenation, solution-phase Ru/Rh/Pd systems at 60 to 200 °C and 0.7 to 15 MPa are disclosed, but catalyst residue removal is essential to preserve aging performance. Reactor intensification via spinning disk hydrogenation offers a lower-pressure alternative, while ion-exchange purification is a practical finishing step for reducing Ru residues. Based on the evidence, the most robust synthesis strategy for low-MW HNBR is controlled NBR precursor preparation followed by simultaneous metathesis/hydrogenation or direct solution hydrogenation with subsequent catalyst removal, aiming for low residual double bonds, Mw below about 250,000 g/mol, and Mooney viscosity in the 1 to 50 range.

5. REFERENCES

- US7666950B2 | Process for preparing hydrogenated nitrile rubbers | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7666950B2/en>
- US8362154B2 | Metathesis of nitrile rubbers in the presence of transition metal complex ... | Not disclosed | US | 2013 | <https://patents.google.com/patent/US8362154B2/en>
- US7951875B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7951875B2/en>

- US20210340285A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | US | 2021 | <https://patents.google.com/patent/US20210340285A1/en>
- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- US9023914B2 | Nitrile rubbers and production thereof in organic solvents | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9023914B2/en>
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- US7846995B2 | ... -containing catalyst residues from optionally hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7846995B2/en>